

OpenShift Disconnected Installation Guide

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Overview

Red Hat OpenShift is a comprehensive Kubernetes platform that simplifies the deployment, management, and scaling of applications, offering support for both containers and virtual machines within a single environment. In Internet-connected environments, OpenShift installation is straightforward with Installer-Provisioned Infrastructure (IPI), a guided, automated method ideal for cloud providers or on-premise setups. However, in disconnected environments, especially those with stringent security requirements, installations become more complex, requiring mirroring of all necessary content locally and tricks to simulate an internet connection for OpenShift's functionality.

Step 1 - Prerequisites for Disconnected Installation

Bastion Host Requirements (connected and disconnected)

- Operating system: RHEL 9
- 2 CPU 8 GB Memory (minimum recommendations)
- Storage minimum of 150 GB of available disk space for image content
- DNS resolvable forward/reverse (disconnected bastion only)
- Install podman and nmstate (disconnected bastion only)
- FIPS enabled if enabling FIPS for OpenShift cluster.
- Set umask to 0022
- Disable fapolicyd

None

```
sudo fips-mode-setup --check
```

```
sudo fips-mode-setup --enable
```

```
umask 0022
```

```
sudo systemctl disable fapolicyd
```

```
sudo dnf install podman nmstate
```

OpenShift Installation Requirements

- **Network** - for bare metal installations it is highly recommended for the ports that connect the bare metal servers to the backend switch to be trunk ports. This allows VMs inside OpenShift to attach to additional VLANs in much the same way as when using VMWare.
 - MAC address of each server will be needed at install time
 - DHCP or Static IP?
 - Bonded network interfaces? Active/Backup or LACP?
 - Identify VLANs?
 - Will a proxy be used, transparent or non-transparent?
 - MTU?
- **NTP** - OpenShift nodes require strict clock synchronization for security handshake protocols and etcd consensus. NTP can be configured through the DHCP Options or added to the agent-config.yaml at install time.
- **DNS**- Each OpenShift host must have a unique IP/DNS resolution
 - DNS forward/reverse entries required for the following entries:
 - api.<cluster_name>.<base_domain>
 - api-int.<cluster_name>.<base_domain>
 - *.apps.<cluster_name>.<base_domain>
 - hostname for each bare metal server
- **Storage** - OpenShift root volumes or ODF (optional) cannot be spinning disks. They must be either SSD or NVME. Recommend RAID 1 for root volume with bare metal installations.
 - Obtain root volume /dev/disk/by-path/<device_name>
 - Example: if installing to /dev/sdd run the following command on the host: ls -l /dev/disk/by-path/ | grep sdd

Step 2 - Download and Configure Software on the Connected Bastion Host

1. Download OpenShift Software

```
None
$ wget
https://mirror.openshift.com/pub/openshift-v4/x86_64/clients/ocp/stable-4.21/openshift-client-linux-amd64-rhel9.tar.gz
$ wget
https://mirror.openshift.com/pub/openshift-v4/x86_64/clients/ocp/stable-4.21/oc-mirror.rhel9.tar.gz
$ wget
https://mirror.openshift.com/pub/openshift-v4/x86_64/clients/ocp/stable-4.21/openshift-install-rhel9-amd64.tar.gz
$ wget
https://developers.redhat.com/content-gateway/file/pub/cgw/mirror-registry/2.0.10/mirror-registry-amd64.tar.gz
```

2. Acquire Pull Secrets

We will need your pull secret to authenticate with the Red Hat servers. In the same folder issue the following commands:

```
None
$ mkdir ~/.docker
$ echo '<YOUR_RED_HAT_PULL_SECRET>' > ~/.docker/config.json
```

3. Configure oc-mirror and download Kubernetes Operators

Red Hat created a tool called oc-mirror to help package up all needed components of openshift. Oc-mirror uses a config file, called imageset.yaml. This config file tells the tool exactly what components to pull down.

The following imageset config file is an example to pull the containers and operators. If you need additional operators that are not included in this file we will explain how to accomplish this in the following section.

None

#This will be the imageconfig.yaml. Cut-and-paste into your command line.

```
$ cat >> imageset.yaml<< EOF
kind: ImageSetConfiguration
apiVersion: mirror.openshift.io/v2alpha1
archiveSize: 4 # Archive chunk size in GB
mirror:
  platform:
    architectures:
      - amd64
    channels:
      - name: stable-4.21
        type: ocp
        minVersion: 4.21.5
        maxVersion: 4.21.5
    graph: true
  operators:
    - catalog: registry.redhat.io/redhat/redhat-operator-index:v4.21
      packages:
        - name: cincinnati-operator
        - name: compliance-operator
        - name: web-terminal
        - name: local-storage-operator
  additionalImages:
    - name: registry.redhat.io/ubi8/ubi:latest
    - name: registry.redhat.io/ubi9/ubi:latest
    - name: registry.redhat.io/rhel8/support-tools
    - name: registry.redhat.io/rhel9/support-tools
EOF
```

To list all operators available in all catalogs, use the following command which will display all operator names for OpenShift 4.21 and save them as a text file in your local directory. Update your imageconfig.yaml with the appropriate operators you want.

None

```
$ for i in $(oc-mirror list operators --catalogs --version=4.21 | grep registry);
do $(oc-mirror list operators --catalog=$i --version=4.21 > $(echo $i | cut -b 27-
| rev | cut -b 7- | rev).txt); done
```

4. Run oc-mirror command - Mirror to Disk

On the connected bastion we will run the oc-mirror command, and pull down all images that are needed to stand up OpenShift + Day 2 operations. This process may take a while. We also need to ensure that umask is set to 0022 on STIG'd RHEL servers to allow the oc-mirror process.

```
None
$ umask 0022
$ oc-mirror --v2 --config ./imageset-config.yaml --cache-dir
/home/allen/.oc-mirror/.cache file:///home/allen/mirror/
```

Step 3 - Transfer Software to Disconnected Network

From here we need to transfer the files over to the disconnected side however you see fit - rsync, scp, sneakernet, burn to a DVD, host it in an apache server, etc. 🏃

We will need to move over the entire directory to include the oc, oc-mirror, mini-quay, etc. binaries, along with the imageset-config and mirror_seq tarballs. Below is the scp command to transfer files over from the connected STIG'ed bastion to the disconnected STIG bastion.

```
None
# scp -r * <username>@<disconnected_bastion_host>:.
$ scp -r * allen@airgap-stig:.
```

Step 4 - Configure Disconnected Bastion Host

On the disconnected side let's unzip our ingredients and check to make sure everything is good to go.

```
None
#Untar our ingredients
$ tar xvf mirror-registry.tar.gz
$ tar xvf oc-mirror.tar.gz
```

```

$ tar xvf openshift-client-linux.tar.gz
$ tar xvf openshift-install-linux.tar.gz

#Move the tools and change ownership make executable
$ sudo cp {mirror-registry,oc,oc-mirror,openshift-install} /usr/local/bin/
$ sudo chown -R $USER
/usr/local/bin/{mirror-registry,oc,oc-mirror,openshift-install}
$ sudo chmod +x /usr/local/bin/{mirror-registry,oc,oc-mirror,openshift-install}
$ sudo restorecon -v
/usr/local/bin/{mirror-registry,oc,oc-mirror,openshift-install}

#We need to disable FApolicyD for the install
$ sudo systemctl stop fapolicyd.service

#The STIG sets max_user_namespaces to 0, preventing rootless containers. Do the
following to enable rootless containers

#Query the current value
$ sysctl -a | grep max_user_namespaces
user.max_user_namespaces = 62372

#Query / change the stored value
$ grep namespaces /etc/sysctl.d/99-sysctl.conf

#Apply the new value
$ sysctl -p /etc/sysctl.d/99-sysctl.conf

# Update /etc/subuid and /etc/subgid with the rootless podman user's id info

```


Step 5 - Installing the Mirror Registry on Bastion Host

As OpenShift is deployed as a set of containers, a registry is necessary to operate properly. In connected environments, OpenShift would use Red Hat's container registry for the initial installation but in disconnected environments this will not be available. Therefore, we need to stand one up to serve this purpose. The installation of the mirror-registry should be done using a general user account

The STIG modifies the user `bashrc` and `profile` to default to `0077`. During the mirroring process to your local registry, we also build your default catalog source. During that process, we need to ensure that the `umask` is set to `0022` so that OpenShift can read those files within the built container. This is necessary because, by default, OpenShift cannot run containers as root for security reasons.

1. Install Quay

None

```
#We had to change the umask from 0077 to 0022 for the quay install, due to the
fact that it always ssh'es into localhost and it get the umask from the default
/etc/bashrc and /etc/profile
sudo sed :%s/077/022/gc /etc/bashrc
suod sed :%s/077/022/gc /etc/profile

#Install the mirror registry (Quay)
#We will need to open up port 8443 for Quay
sudo firewall-cmd --add-port 8443/tcp --permanent
sudo firewall-cmd --reload

#If you have a cert and key that you would like to use instead of having Quay
create a self-signed one please add the following
---
    --sslCert string           The path to the SSL certificate Quay should use
    --sslCheckSkip             Whether or not to check the certificate hostname
against the SERVER_HOSTNAME in config.yaml.
    --sslKey string            The path to the SSL key Quay should use
    -H, --targetHostname string The hostname of the target you wish to install
Quay to. This defaults to $HOST (default
"rhel8-mirror-stig.airgap.dota-lab.iad.redhat.com")
    -u, --targetUsername string The user on the target host which will be used for
SSH. This defaults to $USER (default "allen")
----
```

```
mkdir -p /data/mirror-registry
./mirror-registry install --quayHostname <Full FQDN> --quayRoot
/data/mirror-registry --quayStorage /data/mirror-registry --initUser admin
--initPassword redhat123
```

2. Test if Quay is up and functioning

None

#Copy certificates that created - Only do this if Quay created the certs for you. Quay will create self-signed certs during the install process.

```
sudo cp -v /data/mirror-registry/quay-rootCA/rootCA.pem
/etc/pki/ca-trust/source/anchors/
sudo update-ca-trust
podman login <Full FQDN>:8443
```

3. Upload the Bundled Images into the Registry

None

#Starting the intital image transfer into local Quay - this may take a while
oc-mirror -c imageset-config.yaml --from file:///<path-to-mirror>/ docker:///<Full FQDN>:8443 --v2

#You should notice that your local Quay would start showing images under
https:///<Full FQDN>:8443/repository

Once you recieve message such as images mirrored successfully followed by
Goodbye, thank you for using oc-mirror the disk-to-mirror is complete

Step 6 - Create install-config

Let's create the install-config; the install config is the file that tells CoreOS how to configure itself upon installation. You can find sample install-config.yaml [HERE](#)

1. Grab Pull Secret

First we need to grab the pull secret that will be copied into the config.

```
None
#We will need to get our auth from our local quay registry
$ podman login --authfile config.json <Full FQDN>:8443
$ jq -c -r '.' config.json ###This is your $<PULL_SECRET> that you will need to
copy
```

2. Cut and Paste YAML

Next, cut and paste the following code into the command prompt (first update it with the pull secret and environment specific details):

```
None
$ vi install-config.yaml
---
apiVersion: v1
additionalTrustBundlePolicy: Always
baseDomain: dota-lab.iad.redhat.com <--- #Your domain name
metadata:
  name: ocp4 <--- #Your domain name will get appended to this
compute:
- hyperthreading: Enabled
  name: worker
  replicas: 3
controlPlane:
  hyperthreading: Enabled
  name: master
  replicas: 3
networking:
  clusterNetwork:
  - cidr: 10.128.0.0/14 <--- #IP space for the local network made for within OCP
    hostPrefix: 23
  machineNetwork:
  - cidr: 172.31.255.0/16 <--- #IP space of your physical box
  networkType: OVNKubernetes
  serviceNetwork:
  - 172.30.0.0/16 <--- #IP space for the local network made for within OCP
platform:
  baremetal: <--- For a SNO installs only, replace baremetal with 'none:{}'
```

```

    apiVIPs:
      - 172.31.255.200 <--- #IP reserved for apiVIP, not needed for SNO
    ingressVIPs:
      - 172.31.255.201 <--- #IP reserved for ingressVIP, not needed for SNO
  fips: true
  pullSecret: '$PULLSECRET' <--- #Run this command $jq -c . config.json
  sshKey: '$PUBLIC_RSA_KEY' <--- #Pull your pub key $cat ~/.ssh/id_rsa.pub

#You will need to copy the CA that Quay created from
#/data/mirror-registry/quay-rootCA/rootCA.pem
additionalTrustBundle: |
  -----BEGIN CERTIFICATE-----
  -----END CERTIFICATE-----
# imageDigestSources values are in
# working-dir/cluster-resources/idms-oc-mirror.yaml
imageDigestMirrors:
- mirrors:
  - mirror-registry:8443/openshift/release
    source: quay.io/openshift-release-dev/ocp-v4.0-art-dev
- mirrors:
  - mirror-registry:8443/openshift/release-images
    source: quay.io/openshift-release-dev/ocp-release

```

Step 7 - Create agent-config

Let's also create something called an agent-config.yaml to tell OpenShift how the network is configured:

```
None
$ vi agent-config.yaml
---
#Here is a sample agent-config.yaml for a static IP

apiVersion: v1beta1
kind: AgentConfig
metadata:
  #same name as the cluster in install-config.yaml
  name: stig-sno
rendezvousIP: {{ static_ip_of_box }}
# All fields below are optional
additionalNTPSources: {{ ntp_ip }}
hosts:
  # If a host is listed, then at least one interface
  # needs to be specified.
  - hostname: stig-sno-master-0
    role: master
    rootDeviceHints:
      deviceName: {{ /dev/sda }}
    # MAC Addresses of the NIC
    # interfaces are used to identify the host to apply this configuration to
    interfaces:
      - macAddress: {{ mac_address_1 }}
        name: eth0
      - macAddress: {{ mac_address_2 }}
        name: eth1
      - macAddress: {{ mac_address_3 }}
        name: eth2
      - macAddress: {{ mac_address_4 }}
        name: eth3
    # networkConfig contains the network configuration for the host in NMState format.
    networkConfig:
      interfaces:
        - name: eth0
          type: ethernet
          state: up
          mac-address: {{ mac_address_1 }}
          ipv4:
            enabled: true
            address:
              #static IP of the box
              - ip: {{ static_ip_of_box }}
```

```

        prefix-length: 24
        dhcp: false
- name: eth1
  type: ethernet
  state: down
  mac-address: {{ mac_address_2 }}
  ipv4:
    enable: false
- name: eth2
  type: ethernet
  state: down
  mac-address: {{ mac_address_3 }}
  ipv4:
    enable: false
- name: eth3
  type: ethernet
  state: down
  mac-address: {{ mac_address_4 }}
  ipv4:
    enable: false
dns-resolver:
  config:
    server:
      - {{ router_ip }}
routes:
  config:
    - destination: 0.0.0.0/0
      next-hop-address:{{ gateway_ip }}
      next-hop-interface: eth0
      table-id: 254

```

Step 8 – Create OpenShift Boot ISO

Agent-based installation is a subcommand of the OpenShift installer. It generates a bootable ISO image containing all of the assets required to deploy an OpenShift cluster, with an available OpenShift release image. Create a directory called cluster and copy both install-config.yaml and agent-config.yaml files. The openshift-install will consume the config.yaml files during the agent create image process.

1. Create directory cluster and copy install-config.yaml and agent-config.yaml

```
None
mkdir cluster

cp {install-config.yaml,agent-config.yaml} cluster
```

2. Create Boot ISO Image

Let's create the custom ISO image so we can boot it.

```
None
cd cluster

openshift-install agent create image
```

Step 9 - Load the ISO

Attach media and fire up those server(s). Make note of the URL, username and password to access the GUI upon completion:

None

```
#Go to your IPMI and boot the ISO
```

```
#Attach the media and start the server! - check for UEFI!
```

```
#Back on your disconnected bastion complete the install
```

```
$ openshift-install agent wait-for bootstrap-complete
```

```
$ openshift-install agent wait-for install-complete
```

```
time="2024-05-28T13:44:54-04:00" level=info msg="Install complete!"
```

```
time="2024-05-28T13:44:54-04:00" level=info msg="To access the cluster as the  
system:admin user when using 'oc', run\n    export
```

```
KUBECONFIG=/home/allen/stig/auth/kubeconfig"
```

```
time="2024-05-28T13:44:54-04:00" level=info msg="Access the OpenShift web-console here:  
https://console-openshift-console.apps.stig-sno.dota-lab.iad.redhat.com"
```

```
time="2024-05-28T13:44:54-04:00" level=info msg="Login to the console with user:  
\"kubeadmin\", and password: \"abc-7v9AL-Bd7fI-xyz\""
```


Step 10 – Post Installation Tasks

1. Disabling the default OperatorHub catalog sources

Operator catalogs that source content provided by Red Hat and community projects are configured for OperatorHub by default during an OpenShift Container Platform installation. In a restricted network environment, you must disable the default catalogs as a cluster administrator. You can then configure OperatorHub to use local catalog sources. Additionally we will want to remove the default helmchart and devfile repositories.

None

```
#Disable the sources for the default catalogs by adding disableAllDefaultSources: true to the OperatorHub object:
```

```
export KUBECONFIG=auth/kubeconfig
```

```
oc patch OperatorHub cluster --type json \
  -p '[{"op": "add", "path": "/spec/disableAllDefaultSources", "value": true}]'
```

```
oc patch helmchartrepository openshift-helm-charts --type='merge' -p
'{"spec":{"disabled":true}}'
```

```
oc patch console.operator.openshift.io cluster --type='merge' -p
'{"spec":{"plugins":["dev-console-managed-devfile-catalog-disabled"]}}'
```

2. Updating the Catalog Source

We will need to tell OpenShift that we have a local mirror of all of the operators. Within your bastion host, you could see that oc-mirror created a directory called working-dir/cluster-resources. We will need to apply that directory.

None

```
$ oc apply -f <path-to-mirror>/<path-to-mirror>/working-dir/cluster-resources
```

Troubleshooting

Troubleshooting with common problems when doing bare metal OCP installs.

- DNS issues (make sure your bastion host uses the same DNS as the servers)
- Certs do not link up with NTP, cert is ahead of NTP
 - If needed add NTP ip to agent-config.yaml
- The registry is not up
 - Certs
 - Filesystem issues with STIG
 - Ownership of files
- The pull-secret is not updated with the correct authentication JSON
- Agent-config.yaml with interfaces not called out in both "the upper section and lower section"
- MAC Addresses on agent-config.yaml are correct

References and Documentation

Red Hat Disconnected docs:

https://docs.redhat.com/en/documentation/openshift_container_platform/4.21/html/disconnected_environments/about-disconnected-environments

(Or replace with your OCP Version)

<https://www.redhat.com/en/blog/red-hat-openshift-disconnected-installations>

Red Hat Openshift Docs

https://docs.redhat.com/en/documentation/openshift_container_platform/4.21/

Red Hat Offline Knowledge Portal

https://docs.redhat.com/en/documentation/red_hat_offline_knowledge_portal/1